

Justin P. Bergfield

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Professional Preparation

Educational Training

University of Arizona	Tucson, AZ (USA)	Optical Science, and Physics	Ph.D.	2010
Oxford University	Oxford (England)	Physics	M.Sc.	2008
University of Arizona	Tucson, AZ (USA)	Electrical Engineering	B.S.	2001

Postdoctoral Training

Northwestern University	Evanston, IL (USA)	Prof. Mark Ratner	2012-2014
University of Copenhagen	Copenhagen (Denmark)	Prof. Gemma Solomon	2011-2012
University of California	Irvine, CA (USA)	Prof. Kieron Burke	2010-2011

Appointments

2020-Present	Associate Professor, Department of Physics, Illinois State University
2015-2020	Assistant Professor, Department of Physics, Illinois State University
2014-2015	Research Scientist, Air Force Research Lab
2004-2005	Research Scientist, Raytheon Corporation
2001-2004	Engineer I, Raytheon Corporation

Publications

* Indicates undergraduate authors from my group, ** Indicates graduate authors from my group

- E. Erdogan**, J.P.Bergfield, "The Influence of Quantum Interference and Dephasing on the Local Charge and Heat Flow" (in preparation)
 - E. Erdogan**, J.P.Bergfield, "Quantum Information and Entropy in Molecular Transport Systems: From Coherence to Structure" (in preparation)
 - H. Jilek*, J.P.Bergfield, "Scanning Thermopower Images Quantum Interference" (in preparation)
 - R.X. Bennett, R.X.*, J. Hendrickson, J.P. Bergfield, "The Influence of Off-resonant Transport on the Quantum Enhanced Spin-dependent Thermopower" (in preparation)
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23. J.P. Bergfield, "Identifying Quantum Interference Effects from Joint Conductance-Thermopower Statistics." *Nano Letters*, 24 (16), 12345–12353. DOI: 10.1021/acs.nanolett.4c04439. (2024)
 22. D. Majidi, J.P. Bergfield, V. Maisi, H. Courtois, C.B. Winkelmann, "Heat Transport at the Nanoscale and Ultralow Temperatures—Implications for Quantum Technologies." *Applied Physics Letters*, 124 (14), 140504. DOI: 10.1063/5.0139827. (2024)
 21. R.X. Bennett*, J. Hendrickson, J.P. Bergfield, "Quantum Interference Enhancement of the Spin-Thermopower." *ACS Nano* 2024, 18 (11), 12547–12555. DOI: 10.1021/acs.nano.4c01297. (2024)
 20. M. Baghernejad, C. Van Dyck, J. P. Bergfield, D. Levine, A. Gubicza, J. Tovar, W. Hong, M. Calame, P. Broekmann, "Quantum Interference Enhanced Chemical Responsivity in Single-Molecule Dithienoborepin Junctions ", *Chem. Eur. J.* 25, 15141 (2019).

19. S. Inui*, C.A. Stafford, J.P. Bergfield, "Emergence of Fourier's Law of Heat Transport in Quantum Electron Systems" *ACS Nano* 12 (5), 4304-4311 (2018)
18. J.P. Bergfield, J. Hendrickson, "Signatures of Plexcitonic States in Molecular Electroluminescence," *Nature Scientific Reports* (2018)
17. J. P. Bergfield, H.M Heitzer, C. Van Dyck, T.J. Marks, M. A. Ratner, "Harnessing Quantum Interference in Molecular Dielectric Materials", *ACS Nano* 9 6412-6418 (2015)
16. J.P. Bergfield, M.A. Ratner, C.A. Stafford, M. Di Ventra, "Tunable Quantum Temperature Oscillations in Graphene Nanostructures", *Phys. Rev. B* 91 (2015)
15. J.P. Bergfield, C.A. Stafford, "Thermoelectric Corrections to Quantum Voltage Measurement", *Phys. Rev. B* 90, 235438 (2014)
14. J. Meair, J.P. Bergfield, C.A. Stafford, Ph. Jacquod, "Local Temperature of Out-of-Equilibrium Quantum Electron Systems", *Phys. Rev. B* 90, 035407 (2014)
13. J.P. Bergfield and M.A. Ratner, "Forty Years of Molecular Electronics: Non-equilibrium Heat and Charge Transport at the Nanoscale", *Phys. Status Solidi B*, Review Article, **Cover Article** (2013).
12. J.P. Bergfield, S.M. Story, R.C. Stafford, C.A. Stafford "Probing Maxwell's Demon with a Nanoscale Thermometer", *ACS Nano* 7, 4429-4440 (2013).
11. J.D. Barr, J.P. Bergfield, C.A. Stafford, "Effective Field Theory of Interacting π -Electrons", *Phys. Rev. B* 86, 115403 (2012).
10. ZF. Liu, J.P. Bergfield, K. Burke, C.A. Stafford, "Accuracy of Density Functionals for Molecular Electronics: the Anderson junction", *Phys. Rev. B* 85, 155117 (2012).
9. J.P. Bergfield, ZF. Liu, K. Burke, C.A. Stafford, "Bethe Ansatz Approach to the Kondo Effect within Density-Functional Theory", *Phys. Rev. Lett.* 108, 066801 (2012).
8. J.P. Bergfield, J.D. Barr, C.A. Stafford, "Transmission Eigenvalue Distributions in Highly Conductive Molecular Junctions" *Beilstein J. Nanotechnol.* 3 (2012).
7. G.C. Solomon, J.P. Bergfield, C.A. Stafford, M.A. Ratner, "When 'small' terms matter: coupled interference features in the transport properties cross-conjugated molecules", *Beilstein J. Nanotechnol.* 2, 862 (2011)
6. J.P. Bergfield, G.C. Solomon, C.A. Stafford, M.A. Ratner, "Novel Quantum Interference Effects in Transport through Molecular Radicals", *NanoLetters* 11, 2759-2764 (2011)
5. J.P. Bergfield, J.D. Barr and C.A. Stafford, "The Number of Transmission Channels Through a Single-molecule Junction", *ACS Nano* 5, 2707-2714 (2011)
4. J.P. Bergfield, M. Solis, and C.A. Stafford, "Giant Thermoelectric Effect from Transmission Supernodes", *ACS Nano* 4, 5314-5320 (2010)
3. J.P. Bergfield, P. Jacquod, and C.A. Stafford, "Coherent Suppression of Coulomb-Blockade peaks", *Phys. Rev. B* 82, 205405 (2010)
2. J.P. Bergfield and C. A. Stafford, "Thermoelectric Signatures of Coherent Transport in Single-molecule Heterojunctions", *NanoLetters* 9, 3072-3076 (2009)
1. J.P. Bergfield and C. A. Stafford, "Many-body theory of electronic transport in single-molecule heterojunctions", *Phys. Rev. B* 79, 245125 (2009)

Grants and Awards (Total Amount: \$865,000)

• National Science Foundation 2024 (3yr, \$100K)	Sep 2024
• ISU University Research Grant (\$5k)	Jan 2024
• Air Force Research Lab (AFRL) Grant Awarded (\$60K)	June 2020
• National Science Foundation (NSF-DMR) Award # 1809024 (3yr, \$90K)	July 2019

• Air Force Research Lab (AFRL) Grant Awarded (\$100K)	Sept 2019
• ISU Pre-tenure Undergraduate Research Grant (\$5K)	May 2019
• Air Force Research Lab (AFRL) Grant Awarded (\$100K)	Jan 2018
• ISU Pre-tenure Undergraduate Research Grant (\$5K)	May 2018
• Air Force Research Lab (AFRL) Grant Awarded (\$100K)	Jan 2017
• Air Force Research Lab (AFRL) Grant Awarded (\$100K)	July 2016
• Air Force Research Lab (AFRL) Grant Awarded (\$100K)	Jan 2016
• Northwestern University Nonequilibrium Research Center Fellowship	Jan 2012
• Invited speaker at the American Physical Society (APS) March Meeting	Mar 2010

Selected Administrative and Leadership Accomplishments

• Curriculum Committee	Ongoing
• Engineering Program Coordinator	Ongoing
• Nominations and Awards Committee	Ongoing
• Facilities Committee	Ongoing
• Department Chair Five Review Committee, Member	Spring 2024
• APST Review Committee, Chair	Spring 2023-Fall 2023
• Graduate Program Development Committee, Chair	Spring 2022-Fall 2022
• IAI Compliance Committee, Chair	Spring 2020
• DFSC Committee (Tenure Denial), Member	Fall 2017-Spring 2019

Selected Presentations (*presenter is in bold, *- undergraduate student*)

Note: I was on sabbatical in 2024.

1. J.P. Bergfield, R.X. Bennett, J. Hendrickson "Quantum Interference Enhancement of the Spin-dependent Thermoelectric Response" APS March Meeting Minneapolis (**2024**)
2. J.P. Bergfield, "Quantum Enhancement of the Spin-Thermopower in Single-Molecule Junctions" APS March Meeting Las Vegas (**2023**)
3. R. Wasler*, J.P. Bergfield, "Quantum Thermoelectric Enhancement of Acyclic Cross-conjugated polymers", ISU research symposium (April **2023**)
4. J.P. Bergfield, "Quantum Information of Interference in Quantum Transport" APS March Meeting Chicago (**2022**)
5. **N. Bennett* (senior)**, J.P. Bergfield, "Quantum Interference Enhancement of Spin-Thermopower II", poster uploaded to the online ISU symposium (**2020**)
6. **D. Mueller* (freshman)**, J.P. Bergfield "Developing a Thermoelectric Microscope II", poster uploaded to the online ISU symposium (**2020**)
7. **J.P. Bergfield**, "Harnessing Quantum Correlations with Molecular Electronics," invited speaker at the University of Texas at El Paso (UTEP) (**January 2020**)
8. **N. Bennett* (senior)**, J.P. Bergfield, "Quantum Interference Enhancement of Spin-Thermopower II", poster uploaded to the online ISU symposium (**2020**)
9. **D. Mueller* (freshman)**, J.P. Bergfield "Developing a Thermoelectric Microscope II", poster uploaded to the online ISU symposium (**2020**)

10. **J.P. Bergfield**, "Harnessing Quantum Correlations with Molecular Electronics," invited speaker at the University of Colorado at Colorado Springs (**Online December 2020**)
11. **N. Bennett* (senior)**, J.P. Bergfield, "Quantum Interference Enhancement of Spin-Thermopower", talk given at the ISU symposium on (2019)
12. **C. Sams* (sophomore)**, J.P. Bergfield "Scanning Thermoelectric Microscope Theory", talk given at the ISU symposium on (2019)
13. **D. Mueller* (freshman)**, J.P. Bergfield "Developing a Thermoelectric Microscope", talk given at the NexSTEM conference (2019)
14. **J.P. Bergfield (invited speaker)**, "Emergence of Fourier's law of heat transport in quantum electron systems", talk given at the American Vacuum Society (AVS) Prairie Chapter Symposium on (2019)
15. **N. Bennett* (senior)**, J.P. Bergfield, "Quantum interference enhancement of the spin-thermopower", poster presented at the American Vacuum Society (AVS) Prairie Chapter Symposium (2019)
16. **J.P. Bergfield (invited speaker)**, "Identifying quantum interference effects in Molecular Transport Experiments", Invited to present to the ISU Chemistry Club (2018)
17. **J.P. Bergfield**, "Signatures of Plexcitonic States in Molecular Electroluminescence", APS March Meeting (2018)
18. **C. Sellers* (senior)**, J.P. Bergfield, "Thermoelectric Signature of DNA Nucleotides III", ISU Symposium (2018)
19. **C. Olsen* (senior)**, J.P. Bergfield "Identifying Quantum Interference Effects from thermoelectric transport", ISU Symposium (2018)
20. **G. Mabrey* (senior)**, J.P. Bergfield, "Identifying Quantum Interference Effects from thermoelectric transport II", ISU Symposium (2017)
21. **J. Chaves* (senior)**, J.P. Bergfield, "Quantum thermoelectric enhancement of acyclic cross-conjugated polymers II", ISU Symposium (2017)
22. **C. Sellers* (senior)**, J.P. Bergfield, "Thermoelectric Signature of DNA Nucleotides II", ISU Symposium (2017)
23. **A. Rahe* (senior)**, J.P. Bergfield, "The theory of the scanning thermoelectric microscope", ISU Symposium (2017)
24. **G. Mabrey* (junior)**, J.P. Bergfield, "Identifying Quantum Interference Effects from thermoelectric transport", ISU Symposium (2016)
25. **J. Chaves* (junior)**, J.P. Bergfield, "Quantum thermoelectric enhancement of acyclic cross-conjugated polymers", ISU Symposium (2016)
26. **C. Sellers* (junior)**, J.P. Bergfield, "Thermoelectric Signature of DNA Nucleotides", ISU Symposium (2016)

Other Presentations

- "Molecule-based Material Theory and Design," Air Force Research Lab, (2016)
- **Invited Talk**, "Many-body Theory of Electric and Thermal Transport in Single-molecule Heterojunctions." APS March Meeting, Portland, OR, Mar (2010)
- "Many-body Treatment of Quantum Transport Through Single Molecules." APS Four Corners Meeting, Flagstaff, AZ, Oct (2007). Received the 'Best Student Talk' award for this presentation

Patents

- "High Bandwidth Communication System and Method." US Patent Number US20090245811A1
- "Molecular Thermoelectric Device." International Patent Application WO 2011100438A2 (filed as PCT/US2011/024359, published Aug 18 2011)

Research Experience

- ❖ **Associate Professor of Physics**, Illinois State University Fall 2020-Present
- ❖ **Assistant Professor of Physics**, Illinois State University Fall 2015 – Spring 2020

My research focuses on the nonequilibrium electronic and optical response of nanoscale systems in which quantum coherence and electron–electron correlation are essential. I investigate heat, charge, and spin currents—and their interconversion—in molecular heterojunctions, where one or more molecules bridge bulk materials. These platforms offer a unique opportunity to explore the interplay of quantum duality, strongly correlated matter, nonequilibrium thermodynamics, and information theory. Quantum effects dominate the response in such systems and can be purposefully engineered through molecular design and electrode coupling.

- ❖ **Postdoctoral Fellow with Prof. Mark Ratner**, Northwestern University Spring 2012-Spring 2014

Researched quantum chemical systems in which quantum effects and molecular symmetry can be harnessed to address longstanding technological challenges, for instance, in the development of thermoelectric and “artificial photosynthesis” photovoltaic energy conversion materials.

- ❖ **Postdoctoral Fellow with Prof. Gemma Solomon**, University of Copenhagen Fall 2011-Spring 2012

During my short visit, I investigated the thermoelectric response and phonon heat-transport in cross-conjugated molecule-based devices.

- ❖ **Postdoctoral Fellow with Prof. Kieron Burke**, University of California Irvine Fall 2010-Fall 2011

In order to gain insight into the nature of the exact exchange-correlation functional in density functional theory (DFT), I investigated the Anderson model which can be solved using Bethe Ansatz many-body techniques. I found that the surprising result that the zero-temperature transport is given exactly by DFT, although the spectral function is not. The crossover from mean-field behavior to charge quantization caused by the derivative discontinuity was explored and an accurate approximate functional was developed which may be generalized to larger chemical systems.

- ❖ **PhD thesis research with Prof. Charles Stafford**, University of Arizona Fall 2005-Fall 2010

I developed a state-of-the-art nonequilibrium many-body transport theory which includes correlations on a nanostructure exactly. This non-perturbative theory incorporates interactions not described by other theories but which are necessary, for example, to investigate systems where both Coulomb blockade and resonant tunneling effects contribute (e.g. any small molecular junction). This work was the basis of my invited talk at the American Physical Society March Meeting 2010 and *is the subject of a U.S. patent (US2011/024359).*

Pedagogical Experience

- My teaching emphasizes active learning, accessibility, and the integration of current research into coursework across all levels. In PHY 102, a large-enrollment general education course, I developed both hybrid and asynchronous formats featuring original video lectures, interactive modules, and redesigned assessments. These are paired with real-time formative assessments and **Oxford-style small-group discussion** strategies that foster deep engagement and measurable learning gains.
- **Implemented active learning strategies** in large-enrollment courses, including real-time feedback, discussion-based peer instruction, and group problem solving.
- **Designed and deployed fully hybrid/asynchronous versions of large classes**, with original video lectures, interactive simulations, online discussion forums, and adaptable assessments to improve accessibility and support diverse learning styles.
- **Integrated research-driven projects into upper-level and graduate**, allowing students to explore open-ended questions on topics such as quantum interference, thermoelectrics, and quantum information theory.
- Consistently **updated course content** to reflect emerging developments in physics, including modules on nonequilibrium thermodynamics and quantum information in both undergraduate and graduate settings.
- These innovations have helped bridge the gap between research and instruction, empowering students to think critically, ask deeper questions, and often contribute to ongoing research projects through coursework.

❖ **Assignments as a Professor**, Illinois State University Department of Physics

- *Graduate Statistical Mechanics (PHY425)* Fall 2025-Present
 - Developed a new graduate course in statistical mechanics, introducing ensemble theory, thermodynamic potentials, fluctuations, quantum statistics, and critical phenomena. The course emphasizes the deep connections between statistical mechanics, information theory, and thermodynamics, preparing students for research-level thinking. Class sizes typically range from 10–15 students.
- *Graduate Quantum Mechanics (PHY484)* Fall 2025-Present
 - Developed and taught this graduate course with an emphasis on derivation-based understanding and modern theoretical methods. Students engaged with original problem sets exploring topics such as angular momentum, perturbation theory, path integrals, and Green's functions. The course also incorporated elements of quantum statistical mechanics and connected foundational principles to contemporary research in quantum information and condensed matter physics.
- *Physics Seminar (PHY307, PHY407)* Fall 2015-Present
 - Coordinate and lead the department's professional development seminar for upper-level undergraduates and graduate students, focusing on essential skills for academic and professional success in physics. Topics include crafting effective CVs and personal statements, designing scientific posters, delivering oral presentations, and navigating research careers. The course features peer feedback, faculty panels, and invited talks, fostering a supportive and skill-building environment. Typical enrollment ranges from 10–20 students per semester.

- *From Atoms to Galaxies (PHY102)* Fall 2015-Present
 - Designed and taught a dynamic general education physics course for non-science majors, covering a broad sweep of topics from Newtonian mechanics and electromagnetism to relativity, quantum theory, and thermodynamics. With typical enrollments of ~180 students, the course emphasizes scientific reasoning, conceptual understanding, and real-world relevance, supported by custom lecture slides, live demonstrations, and active learning strategies. Taught a total of 7 times, reaching thousands of students across disciplines.

- *Thermal Physics (PHY325)* Fall 2015-Present
 - Developed and teach a core upper-level course for physics majors focusing on classical thermodynamics and statistical mechanics, including heat engines, entropy, partition functions, and quantum statistics. Emphasis is placed on bridging microscopic and macroscopic perspectives through hands-on derivations, conceptual insight, and applications to real systems. Typical class size is ~15 students.

- *Solid State Physics (PHY355)* Fall 2015-Present
 - Created and implemented a solid-state physics course that forms the foundation of ISU's solid-state track, covering crystal structure, phonons, electronic band theory, semiconductors, and device physics including transistor operation. The course blends theoretical concepts with practical applications and includes problem-solving, computational tools, and experimental relevance. Typical enrollment is ~15 students.

- *Quantum Mechanics (PHY284)* Fall 2015-Present
 - Designed this sophomore-level introduction to quantum mechanics to develop students' foundational understanding of wave-particle duality, Schrödinger's equation, probability amplitudes, and operator formalism. The course builds strong conceptual and mathematical fluency through analytical problem sets and guided exploration of core physical systems. Typical enrollment is ~20 students.

- ❖ **Assignments as an Instructor, University of Arizona**
 - *Introduction to Physics II (PHYS 103)* Summer 2010
 - Prepared and conducted daily two-hour lectures, discussion sections, demonstrations, homework and exams for a 3-unit physics course directed toward non-physics majors on electromagnetism and modern physics for a class of around 60 students. Managed a team of two teaching assistants and held daily office hours.

- ❖ **Assignments as a Graduate Teaching Assistantships, University of Arizona**
 - *Undergraduate Electromagnetism Lab (PHYS 182), Department of Physics* Fall 2009
 - Directed three lab sections, 20-30 undergraduate students each (nine hours per week). Responsible for laboratory lectures, running lab, office hours and grading reports.

 - *Imaging Physics and Devices (OPTI 537), College of Optical Science* Fall 2008
 - Led recitation section, graded homework, held office hours and gave several lectures for a class of around 30 graduate students.

 - *Holography and Diffractive Optics (OPTI 527), College of Optical Science* Fall 2008
 - Prepared and ran holography labs, graded homework, and gave several lectures for a class of around 40 graduate students.

- *Semiconductor Physics and Lasers (OPTI 240)*, College of Optical Science Spring 2008
 - Led recitation section, graded homework, held office hours and gave several lectures for a class of around 60 undergraduate students.
- *Electromagnetic Waves (OPTI 501)*, College of Optical Science Fall 2007
 - Led recitation section and graded homework and exams for a class of 150 graduate students.
 - Received the professor-nominated 'Rolyn Optics Outstanding Graduate Teaching Award' for this appointment based on student feedback.
- *Advanced Fourier Methods (OPTI 474)*, College of Optical Science Spring 2005
 - Responsible for review sessions, homework and occasional lectures for a class of 55 undergraduate students.
- *Lasers and Photonics (OPTI 370)*, College of Optical Science Fall 2004
 - I led the recitation section, graded homework and occasionally lectured for a class of around 50 undergraduate students.
 - Received the professor-nominated 'Rolyn Optics Teaching Excellence Award' for this appointment based on student feedback.
- ❖ **Assignments as a Graduate Teaching Assistant**, University of Oxford
 - *Electromagnetism Tutorials*, St. Hilda's and St. Hugh's Colleges Hilary Term 2007
 - Conducted five tutorials a week with 10 students on the topic of electromagnetism.
 - Responsibilities included writing lectures, grading homework and preparing additional study materials.
 - *Undergraduate Optics Lab*, Clarendon Laboratory Michaelmas Term 2006
 - Supervised an undergraduate optics lab three days a week.

Industry Experience

- ❖ **Research Scientist**, Air Force Research Laboratory – Sensors Directorate Spring 2014-Fall 2015

Pursued multiple theoretical and computational projects in quantum optics and nanophotonics, with a focus on strong electron-electron correlations and light-matter coupling in nanoscale and quantum-engineered systems. Collaborated with AFRL physicist Joshua R. Hendrickson on research into epsilon-near-zero metamaterials and hybrid photonic-electronic platforms, contributing to AFRL's objectives in infrared and THz sensing technologies. This work led to several peer-reviewed publications:

- ❖ **Research Scientist**, Raytheon Corporation May 2005-December 2005

Directed an R&D feasibility study on THz radiation generation via the Smith-Purcell effect in negative index metamaterials. This early-stage investigation explored novel mechanisms for compact, efficient THz sources. *This work is also the basis for our patent (US20090245811A1): "Method and Apparatus for Smith-Purcell Radiation Generation Using Negative Index Materials"*

- ❖ **Intern (1998-2001), Engineer I (2001-2004)**, Raytheon Corporation May 1998-May 2001

Began working as a summer intern while completing my undergraduate and early graduate studies, transitioning to full-time engineer and later research scientist. Over this time, I contributed to a broad range of advanced technology and R&D efforts within Raytheon's defense, optics and sensors programs.

- Developed and maintained high-energy laser simulation codes in C++, including integration with large commercial finite-element and modeling packages such as PATRAN.
- Participated in classified and unclassified Advanced Programs efforts focused on directed energy systems, optical propagation modeling, and defense technology prototyping.
- Conducted original research into photonic crystal devices engineered for modified thermionic emission, aiming to produce efficient, tunable infrared radiation. Designed and simulated photonic crystal structures that were later fabricated in-house based on our modeling results.